10, 18

a,

```
## Mother some college percentage: 12.14953
```

```
## Father some college percentage: 11.68224
```

```
table(dat$mothercoll)
```

```
##
##   0   1
## 376  52
```

← mother

1 have college education

0 don't have college education

```
table(dat$fathercoll)
```

```
##
##   0   1
## 378  50
```

← father

b,

```
# (b)
cor(dat[, c("educ", "mothercoll", "fathercoll")])
```

```
##                 educ mothercoll fathercoll
## educ       1.0000000  0.3594705  0.3984962
## mothercoll 0.3594705  1.0000000  0.3545709
## fathercoll 0.3984962  0.3545709  1.0000000
```

cov(educ, mothercoll) = 0,359

cov(educ, fathercoll) = 0,398

=> 和EDUC有不小的相關性

∴ 可能是 good IV

```
# optional comparison
cor(dat[, c("educ", "mothereduc", "fathereduc")])
```

```
##                 educ mothereduc fathereduc
## educ       1.0000000  0.3870198  0.4154030
## mothereduc 0.3870198  1.0000000  0.5540632
## fathereduc 0.4154030  0.5540632  1.0000000
```

有上過大學可能對子女教育較為重視

∴ (coll better than educ)

c,

```
##              2.5 %     97.5 %
## educ -0.001219763 0.1532557
```

d,

```
## First-stage F in part (d): 63.56307
```

since F is very large

=> MOTHERCOLL is strong instrument

e,

```
##              2.5 %     97.5 %
## educ 0.02751845 0.1481769
```

it is narrower than CI in (c)

f,

```
##
## Linear hypothesis test:
## mothercoll = 0
## fathercoll = 0
##
## Model 1: restricted model
## Model 2: educ ~ mothercoll + fathercoll + exper + I(exper^2)
##
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    425 2219.2
## 2    423 1748.3  2    470.88 56.963 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

∵ F = 56,963 is large

Hence it is adequately strong instruments.

g,

```
## Diagnostic tests:
##                  df1 df2 statistic p-value
## Weak instruments   2 423    56.963  <2e-16 ***
## Wu-Hausman         1 423     0.519   0.472
## Sargan             1  NA     0.238   0.626
```

since $NR^2 = 0,238$, and p-value = 0,626

=> we do not reject validity of surplus instrument.

10, 20

a,

```
## OLS beta = 1.20184
```

$\hat{\beta} > 1 \Rightarrow$ it is more risky than the market portfolio

b,

```
##
## Call:
## lm(formula = MKT_RF ~ RANK, data = dat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.110497 -0.006308  0.001497  0.009433  0.029513
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -7.903e-02  2.195e-03   -36.0   <2e-16 ***
## RANK         9.067e-04  2.104e-05    43.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01467 on 178 degrees of freedom
## Multiple R-squared:  0.9126, Adjusted R-squared:  0.9121
## F-statistic:  1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

$\rightarrow \widehat{MKT\text{-}RF} = -0{,}0793 + 0{,}0009067 RANK$

```
## First-stage F using RANK = 1857.587
```

```
## First-stage R2 using RANK = 0.9125559
```

$\rightarrow$ Since F is large

$\therefore$ RANK is good IV for MKT-RF

C,

```
## Hausman p-value using RANK = 0.04278906
```

$\because 0{,}0427 > 0{,}01$

$\Rightarrow$ we do not reject that the market return is exogenous d,

(a)

```
##                    2.5 %     97.5 %
## (Intercept) -0.008661409 0.01516061
## MKT_RF        0.960788169 1.44289134
```

```
## OLS beta = 1.20184
```

(d)

```
##                    2.5 %     97.5 %
## (Intercept) -0.008827134 0.01486322
## MKT_RF        1.027421458 1.52921503
```

```
## IV beta using RANK = 1.278318
```

$\hat{\beta}_{IV} > \hat{\beta}_{OLS}$, it is satisfied the expect about measurement error.

e,

```
## First-stage R2 using RANK and POS = 0.9148844
```

```
##
## Linear hypothesis test:
## RANK = 0
## POS = 0
##
## Model 1: restricted model
## Model 2: MKT_RF ~ RANK + POS
##
##   Res.Df     RSS Df Sum of Sq      F    Pr(>F)
## 1    179 0.43784
## 2    177 0.03727  2   0.40057 951.26 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$\because$ F = 951.26 is large, we conclude that Rank, POS is adequately strong IV f,

```
## Hausman p-value using RANK and POS = 0.02874032
```

> 0,01

∵ we do not reject $H_0: \delta = 0$ st, it is exogenous

g,

(g)

```
##                   2.5 %     97.5 %
## (Intercept) -0.008827134 0.01486322
## MKT_RF       1.027421458 1.52921503
```

```
## IV beta using RANK and POS = 1.283118
```

(a)

```
##                   2.5 %     97.5 %
## (Intercept) -0.008661409 0.01516061
## MKT_RF       0.960788169 1.44289134
```

```
## OLS beta = 1.20184
```

$\hat{\beta}_{IV} > \hat{\beta}_{OLS}$ ⇒ 符合預期 (ols 會低估 β, if has measurement error)

h,

```
## Sargan statistic = 0.5584634
```

```
## Sargan p-value = 0.45488
```

> 0,05 ⇒ we do not reject surplus IV validity at 5% Level